Midterm Practice

1 Proofs

Problem 1

Show that the square of an even number is also an even number using both:

- A Direct Proof
- A Proof by Contradiction

Solution

A Direct Proof *Proof.* Let n be an even number. In other words, let n be an integer that can be written as $2k$, with $k \in \mathbb{Z}$.

Squaring this number, we have $n^2 = (2k)^2 = 4k^2$ We can simplify this expression to $2 * (2k^2)$. Since k is an integer, we know that $2k^2$ is an integer. If we let a new integer $m = 2k^2$, then we can say that $n^2 = 2 * m$, which demonstrates that n^2 is two times some integer m, proving that n^2 must be even. □

Proof by Contradiction *Proof.* Assume for the sake of contradiction that there exsists a square of an even number that is not even (is odd). Let $2k$ be any even number. By assumption, there must be at least one even number $2k$ for which the square is odd:

- $(2k)^2 = 2j + 1$ where 2j+1 is an odd number
- $2(2k^2 - j) = 1$ follows
- $2k^2 - j$ must be an integer for integers k, j
- $2k^2 - j = 1/2$

As one half is not an integer, we've reached a contradiction. This means the square of an even is never odd, and thereby always even. □

Problem 2

Prove that $\forall n \in \mathbb{Z},\ n^2 + 3n - 5$ is odd by dividing into cases.

Solution

We begin by proving that an even number times an odd number is even. An even number takes the form $2k$ and an odd number takes the form $2k+1$ for an integer $k \in \mathbb{Z}$.

Consider an even number $a = 2i$ for some $i \in \mathbb{Z}$ and an odd number $b = 2j+1$ for some $j \in \mathbb{Z}$. The product $ab = (2i)(2j+1) = 4ij + 2i = 2(2ij+i)$. Thus, since $(2ij+i) \in \mathbb{Z}$, ab is even.

We then note that

$$n^2 + 3n - 5 = (n+1)(n+2) + 3$$

Now, there are two cases to consider:

n **is even:** If n is even, then n can be expressed in the form $2k$ for some integer $k \in \mathbb{Z}$. Thus, $n+1 = 2k+1$ which by definition implies that $n+1$ is odd. Furthermore, $n+2 = 2k+2 = 2(k+1)$ which by definition implies that $n+2$ is even. Hence, by the proof given at the start, $(n+1)(n+2)$ is even, meaning it has a factor of two. We conclude that n^2+3n-5 is odd since

$$(n+1)(n+2) + 3 = 2\left(\frac{(n+1)(n+2)}{2} + 1\right) + 1$$

n **is odd:** If n is odd, then n can be expressed in the form $2k+1$ for some integer $k \in \mathbb{Z}$. Thus, $n+1 = 2k+2 = 2(k+1)$ which by definition implies that $n+1$ is even. Furthermore, $n+2 = 2k+3 = 2(k+1)+1$ which by definition implies that $n+2$ is odd. Hence, by the proof given at the start, $(n+1)(n+2)$ is even, meaning it has a factor of two. We conclude that n^2+3n-5 is odd since

$$(n+1)(n+2) + 3 = 2\left(\frac{(n+1)(n+2)}{2} + 1\right) + 1$$

Since both cases have been shown to be odd, we have successfully demonstrated that $\forall n \in \mathbb{Z},\ n^2+3n-5$ is odd.

2 Logic/Circuits

Problem 1

a. Suppose p and q are propositions such that p IMPLIES q is False. Determine the values of:

(i) (NOT p) IMPLIES q

(ii) p OR q

(iii) q IMPLIES p

b. You are now going to design a circuit that takes as input two 1-bit binary numbers A and B and outputs whether or not $A > B$, $A < B$, or $A = B$. Namely, the circuit should have two inputs, the bits A and B, and three outputs G, E, and L. For any input, exactly one of G (greater), E (equal), or L (less) should be 1. Specifically, G corresponds to $A > B$.

(i) Write out a truth table equivalent to this circuit.

(ii) Draw your circuit. Please use only AND, OR, and NOT gates with at most two inputs per gate. Note that we care only about the correctness of the circuit, not its complexity.

Be sure to explain how your circuit works.

Solution

a. Because p IMPLIES q is False, p must be True and q must be False.

   i. (NOT p) IMPLIES q is True

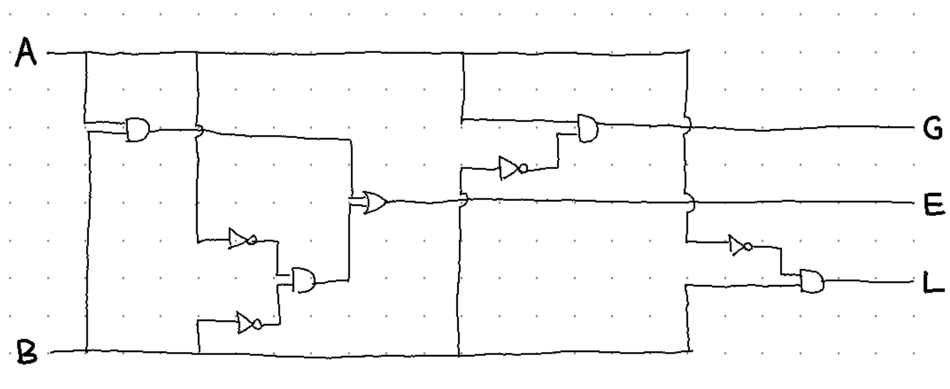
   ii. p OR q is True

   iii. q IMPLIES p is True

b. i.

A	B	G	E	L
T	T	F	T	F
T	F	T	F	F
F	T	F	F	T
F	F	F	T	F

   ii. This is one possible circuit:



The first AND gate takes A and B as inputs to check whether they are both true: A AND B. The second AND gate takes the negations of A and B to check whether they are both false: (NOT A) AND (NOT B). The outputs of those AND gate are then ORed together, so E is true if and only if A and B are both true or A and B are both false, which is the same as saying $A = B$.

From the above truth table, we see that G is only true if A is true and B is false, so we AND together A and the negation of B to get G. L is similar to G except that B has to be true and A has to be false.

Problem 2

a. Add parentheses to the following expressions to make them true (1 represents true, and 0 represents false). Note also that there is no implicit ordering; that is, all ordering comes from your parentheses. State explicitly any assumptions you are making about the order of operations.

i. 0 AND 1 OR 1 IMPLIES 1 AND 1 AND 1 OR 0

ii. 0 OR 0 AND 1 AND 0 AND 1 AND 1 OR 1

iii. 0 AND 1 OR 1 IFF 0 IMPLIES 0

iv. 0 OR 1 IMPLIES 0 AND 0 IMPLIES 1

b. Prove the following logical equivalences. You may use truth tables, or logical rewrite rules (both will be included in the answers).

i. NOT (p OR (NOT p AND q) $\equiv$ NOT p AND NOT q

ii. (p IMPLIES r) AND (q IMPLIES r) $\equiv$ (p OR q) IMPLIES r

c. Prove that q AND NOT (p IMPLIES q) is unsatisfiable.

d. Prove that (p AND q) IMPLIES (p OR q) is valid.

Solution

a. There are many possible solutions for each problem. **Note**: Since this question asked you to add parentheses, none of the sub-expression of operations should have been left ambigious, even if the expression evaluates to true under all possible placements of parentheses.

i. Example: $((0 \wedge 1) \vee 1) \Rightarrow ((1 \wedge 1) \wedge (1 \vee 0))$
All syntactically correct expressions evaluated to true if "$\Rightarrow$" was the highest-order term (i.e. "$\Rightarrow$" separated the expression into two sub-expressions), regardless of the other parentheses. There are other possible true answers.

ii. Example: $(((0 \vee 0) \wedge (1 \wedge 0)) \wedge (1 \wedge 1)) \vee 1$
All syntactically correct expressions evaluated to true, regardless of parentheses.

iii. Example: $((0 \wedge 1) \vee 1) \Leftrightarrow (0 \Rightarrow 0)$
All syntactically correct expressions evaluated to true, regardless of parentheses.

iv. Example: $((0 \vee 1) \Rightarrow (0 \wedge 0)) \Rightarrow 1$

All syntactically correct expressions evaluated to true if and only if there were not parentheses arround the last two terms; that is, if and only if the expression did not include $(0 \Rightarrow 1)$.

b. We could do these problem in two ways: with logical rewrite rules or truth tables. Both are shown below. Note that if you use the truth table method, you should show your work (the intermediate columns).

i. Truth table method:

p	q	$\neg p$	$\neg q$	$\neg p \wedge q$	$p \vee (\neg p \wedge q)$	$\neg(p \vee (\neg p \wedge q))$	$\neg p \wedge \neg q$
T	T	F	F	F	T	F	F
T	F	F	T	F	T	F	F
F	T	T	F	T	T	F	F
F	F	T	T	F	F	T	T

Logical rewrite rules:

$$\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{DeMorgan's Law}\\
&\equiv \neg p \wedge (p \vee \neg q) && \text{DeMorgan's Law}\\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{distributive law}\\
&\equiv \mathrm{F} \vee (\neg p \wedge \neg q) && \text{universal bounds}\\
&\equiv \neg p \wedge \neg q && \text{identity law}
\end{aligned}$$

ii. Truth table method:

p	q	r	$p \Rightarrow r$	$q \Rightarrow r$	$(p \Rightarrow r) \wedge (p \Rightarrow r)$	$p \vee q$	$(p \vee q) \Rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	F
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

Logical rewrite rules:

$$\begin{aligned}
(p \Rightarrow r) \wedge (q \Rightarrow r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{def. of } \Rightarrow\\
&\equiv (\neg p \wedge (\neg q \vee r)) \vee (r \wedge (\neg q \vee r)) && \text{distributive law}\\
&\equiv ((\neg p \wedge \neg q) \vee (\neg p \wedge r)) \vee ((r \wedge \neg q) \vee (r \wedge r)) && \text{distributive law}\\
&\equiv ((\neg p \wedge \neg q) \vee (\neg p \wedge r)) \vee ((r \wedge \neg q) \vee r) && \text{idempotent law}\\
&\equiv ((\neg p \wedge \neg q) \vee (\neg p \wedge r)) \vee r && \text{absorption law}
\end{aligned}$$

$\equiv (\neg p \land \neg q) \lor ((\neg p \land r) \lor r)$	associative law
$\equiv (\neg p \land \neg q) \lor r$	absorption law
$\equiv \neg(p \lor q) \lor r$	DeMorgan's law
$\equiv (p \lor q) \Rightarrow r$	def. of $\Rightarrow$

c. For $q \land \neg(p \Rightarrow q)$ to be true, there has to be one instance in which both q and $\neg(p \Rightarrow q)$ are true. The following truth table shows that this is not true for any values combinations of p and q.

p	q	$p \Rightarrow q$	$\neg(p \Rightarrow q)$	q	$q \land \neg(p \Rightarrow q)$
T	T	T	F	T	F
T	F	F	T	F	F
F	T	T	F	T	F
F	F	T	F	F	F

This problem could also be done with logical rewrite rules.

d. To prove that something is valid, we need to show it is true in all rows.

p	q	$p \land q$	$p \lor q$	$(p \land q) \Rightarrow (p \lor q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

This problem could also be done with logical rewrite rules.

3 Set Theory

Problem 1

Determine whether each of the following statements is true or false, and explain why.

a. The powerset of the empty set has the empty set as a member.

b. The empty set is an element of every set.

c. The empty set is a subset of any set that does not have the empty set as a member.

d. Any set that has the empty set as a member must be the empty set.

e. The set containing the set containing the set containing the empty set has a cardinality of zero (here, A contains B means $B \in A$).

f. The set of all empty sets is the same as the powerset of the empty set.

Solution

a. True, as empty set is subset of any set, including itself.

b. False. Not an element of the empty set.

c. True, but true in general as well.

d. False. If it has members, it's not the empty set.

e. False. Cardinality of 1.

f. True (there's only one empty set, so this is a 1 element set).

Problem 2

a. Disprove the following claim:

   For any two sets finite A and B, there are the same number of elements in $\mathcal{P}(A \times B)$ and $\mathcal{P}(A) \times \mathcal{P}(B)$.

b. Consider two finite, *non-empty* sets A and B. Suppose $\mathcal{P}(A \times B)$ and $\mathcal{P}(A) \times \mathcal{P}(B)$ were equal in size. What must be the size of A? What must be size of B? Explain your answer.

Solution

a. Call the cardinality of set A, n, and the cardinality of set B, m.

In $A \times B$, each element of A is related to each element of B. Thus, for each element a in A, there are m pairs that contain a in $A \times B$. There are n total elements in A, so the cardinality of $A \times B$ is mn.

We know that given some set of size k, the powerset of our set is of size 2^k.

The size of $A \times B$ is mn. Therefore, the size of $\mathcal{P}(A \times B)$ is 2^{mn}.

The size of A is n, and thus, the size of $\mathcal{P}(A)$ is 2^n.

The size of B is m, and thus, the size of $\mathcal{P}(B)$ is 2^m.

In $\mathcal{P}(A) \times \mathcal{P}(B)$, each element of $\mathcal{P}(A)$ is related to each element of $\mathcal{P}(B)$. Thus, for each element a in $\mathcal{P}(A)$, there are 2^m pairs that contain a in $\mathcal{P}(A) \times \mathcal{P}(B)$. There are 2^m total elements in $\mathcal{P}(A)$, so the cardinality of $\mathcal{P}(A) \times \mathcal{P}(B)$ is $2^m \cdot 2^n$. We can re-express this as 2^{m+n}.

We have that the size of $\mathcal{P}(A \times B)$ is 2^{mn}, and the size of $\mathcal{P}(A) \times \mathcal{P}(B)$ is 2^{m+n}. 2^{mn} is not equal to 2^{m+n}. Thus, there are not the same number of elements in $\mathcal{P}(A \times B)$ and $\mathcal{P}(A) \times \mathcal{P}(B)$.

b. In part a, we showed that the size of $\mathcal{P}(A \times B)$ is 2^{mn}, and the size of $\mathcal{P}(A) \times \mathcal{P}(B)$ is 2^{m+n}.

Now, A and B are non-empty, and thus m and n must be positive integers. 2^{mn} is equal to 2^{m+n} when $m = n = 2$, as mn is equal to $m + n$ when $m = n = 2$. There are no other positive integers m and n such that mn is equal to $m + n$.

Thus, the cardinality of both A and B must be 2.

NOTE: Any counterexample suffices to prove part a. However, if the student provides only a counterexample for part a, they must do what we do here for part a in part b.

4 Relations

Problem 1

a. Prove or disprove that a relation that is reflexive and symmetric is necessarily transitive.

b. Prove or disprove that a relation that is reflexive and transitive is necessarily symmetric.

c. Prove or disprove that a relation that is symmetric and transitive is necessarily reflexive.

Solution

Disproving all by counterexample. Let S be a set, and let R be a relation on S:

a. $S = \{a, b, c, d\}$ and $R = \{(a, a), (a, c), (b, b), (b, c), (c, a), (c, b), (c, c), (d, d)\}$.

We will show that R is reflexive and symmetric, but not transitive.

R is reflexive because for all elements in S, $(a, a), (b, b), (c, c), (d, d) \in R$.

R is symmetric because for all elements in R, the reversed tuple is also in R: that is, $(a, c), (c, a) \in R$, $(b, c), (c, b) \in R$, and $(a, a), (b, b), (c, c), (d, d) \in R$.

R is not transitive because $(a, c), (c, b) \in R$ but $(a, b) \notin R$.

b. $S = \mathbb{Z}$ and $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x \leq y\}$.

We will show that R is reflexive and transitive, but not symmetric.

R is reflexive, because for all $a \in S$, $a \leq a$ and $a \leq a$ hold. So $(a, a) \in R$.

R is transitive, because if $a, b, c \in S$ and $(a, b) \in R$ and $(b, c) \in R$, then $a \leq b$ and $b \leq c$. So $a \leq c$. Thus, $(a, c) \in R$.

R is not symmetric, because $1 \leq 2$, but $2 \not\leq 1$.

(In fact, R is antisymmetric, thereby making it a partial order. We leave as an exercise to the reader.)

c. $S = \{a, b, c\}$ and $R = \{\}$.

We will show that R is symmetric and transitive, but not reflexive.

R is vacuously symmetric. That is, because R is empty, it is never the case that $(x, y) \in R$ for any $x, y \in S$. Therefore, symmetry is never contradicted.

R is transitive by an analogous argument.

R is not reflexive, because it is not the case that for every element s in S, $(s, s) \in R$. In fact, this does not hold for any $s \in S$.

Note: R does not have to be nonempty to make a counterexample. For example, another counterexample is $A = \{a, b, c\}$ and $R = \{(a, a), (a, b), (b, a), (b, b)\}$.

Problem 2

Let $S = \{0, 1\}$, $T = \{t | t \subseteq S \times S\}$, and R be the set of all possible functions from S to S.

a. Can an injection from T to R exist? If so, give one such injection and prove that this mapping is indeed injective. If not, prove why such a mapping cannot exist.

b. Can a surjection from T to R exist? If so, give one such surjection and prove that this mapping is indeed surjective. If not, prove why such a mapping cannot exist.

c. Can a bijection from T to R exist? If so, why? If not, why not?

Solution

a. An injection from T to R cannot exist. The cardinality of T is $2^{|S|^2}$, or 2^{2^2}, or 16. The cardinality of R is $|S|^{|S|}$, or 2^2, or 4. Because the codomain is smaller than the domain, each element of T cannot be mapped to a unique element of R, so no injection can exist.

b. A surjection from T to R can exist because the cardinality of the domain is larger than the cardinality of the codomain.

   Let A be a function from S to S where 0 maps to 1 and 1 maps to 1. Let B be a function from S to S where 0 maps to 0 and 1 maps to 0. Let C be a function from S to S where 0 maps to 0 and 1 maps to 1. Let D be a function from S to S where 0 maps to 1 and 1 maps to 0. R is the set $\{A, B, C, D\}$.

   T is the power set of the Cartesian product of S. For our surjection, we will map $\{\}$ to A, $\{(0, 0)\}$ to B, $\{(1, 1)\}$ to C, and all other elements of T to D.

   Proof. We will show that all elements of our codomain, R, are mapped to by our function, meaning that A, B, C, and D are all covered. A is mapped to by $\{\}$, B is mapped to by $\{(0, 0)\}$, C is mapped to by $\{(1, 1)\}$, and B is mapped to by $\{(0, 1)\}$. Because for each element of our codomain we have given an element that maps to it, we have proven that the function is surjective. $\square$

c. A bijection from T to R cannot exist because for a bijection to exist, an injection must also exist, but we proved above that an injection cannot exist from T to R.

5 Induction

Problem 1

Suppose we have a sequence defined as follows:

$$a_1 = 1$$
$$a_2 = 3$$
$$a_k = a_{k-2} + 2a_{k-1}$$

Prove by induction that a_n is odd for all positive integers n.

Solution

Lemma: $\forall x, y$ where x is even and y is odd, $x + y$ is odd.

Proof: x, being even, can be written as $2a$ for some integer a. y, being odd, can be written as $2b + 1$ for some integer b.

So $x + y = 2a + 2b + 1 = 2(a + b) + 1 = 2c + 1$, where $c = a + b$. Any number that can be represented as $2c + 1$ for some integer c is odd, so $x + y$ is odd.

We're going to use strong induction. Let $P(n)$ be the predicate that a_m is odd for all $1 \leq m < n$.

Base Case: We know $P(3)$ is true, since a_1 and a_2 are both odd. And $P(2)$ is also true.

Inductive Hypothesis: Assume $P(k)$ for some integer $k \geq 3$.

Inductive Step: From the Inductive Hypothesis, we can say a_{k-1} and a_{k-2} are both odd. We want to show that a_k is also odd. $a_k = a_{k-2} + 2 * a_{k-1}$. Let's say $x = 2 * a_{k-1}$ and $y = a_{k-2}$. Since x is 2 times an integer, it must be even, and we already said y must be odd. Since $a_k = x + y$, we know from our lemma that a_k is odd.

Conclusion: Since $P(n)$ holds for all positive integers $n > 1$, a_k must be odd for all positive integers.

Problem 2

Use mathematical induction to prove the following generalization of one of DeMorgan's laws:

$$\overline{\bigcap_{j=1}^{n} A_j} = \bigcup_{j=1}^{n} \overline{A_j},$$

whenever $A_1, A_2, ..., A_n$ are subsets of a universal set U and $n \geq 2$. To give you an idea of what this notation means, note that:

$$\bigcap_{j=1}^{3} A_j = A_1 \cap A_2 \cap A_3.$$

Note: You must prove DeMorgan's Law for the base case.

Solution

Proof. **Proposition:**

The proposition $P(n)$ we wish to prove by induction is

$$\overline{\bigcap_{j=1}^{n} A_j} = \bigcup_{j=1}^{n} \overline{A_j}$$

whenever $A_1, A_2, ..., A_n$ are subsets of a universal set U and $n \geq 2$.

Base Case: Consider the case $n = 2$.

We want to show our argument holds for $n = 2$. So, we want to show that $\overline{\bigcap_{j=1}^{2} A_j} = \bigcup_{j=1}^{2} \overline{A_j}$. This simplifies to showing that $\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$. We will prove this using the set-element method.

Sub-proof:

Proof. We want to show that $\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$. So, we will show that $\overline{A_1 \cap A_2} \subseteq \overline{A_1} \cup \overline{A_2}$, and then we will show that $\overline{A_1} \cup \overline{A_2} \subseteq \overline{A_1 \cap A_2}$.

First, consider $x \in \overline{A_1 \cap A_2}$. We know that x is not $A_1 \cap A_2$. In other words, x is not in both A_1 and A_2. That means that either x is not in A_1, or x is not in A_2. Written in notation, $x \notin A_1$ or $x \notin A_2$, which we can rewrite as $x \in \overline{A_1}$ or $x \in \overline{A_2}$. Finally, by the definition of set union, we can rewrite this as $x \in \overline{A_1} \cup \overline{A_2}$.

Therefore, since $x \in \overline{A_1 \cap A_2}$ implies that $x \in \overline{A_1} \cup \overline{A_2}$ for all x, we can conclude that $\overline{A_1 \cap A_2} \subseteq \overline{A_1} \cup \overline{A_2}$.

Next, consider $x \in \overline{A_1} \cup \overline{A_2}$. By the definition of set union, we know that $x \in \overline{A_1}$ or $x \in \overline{A_2}$, which we can rewrite as $x \notin A_1$ or $x \notin A_2$. That means that x cannot be in both A_1 and A_2, which implies by the definition of set intersection that x is not in $A_1 \cap A_2$. Therefore, $x \in \overline{A_1 \cap A_2}$.

So, since $x \in \overline{A_1} \cup \overline{A_2}$ implies that $x \in \overline{A_1 \cap A_2}$ for all x, we can conclude that $\overline{A_1} \cup \overline{A_2} \subseteq \overline{A_1 \cap A_2}$.

Finally, since $\overline{A_1 \cap A_2} \subseteq \overline{A_1} \cup \overline{A_2}$ and $\overline{A_1} \cup \overline{A_2} \subseteq \overline{A_1 \cap A_2}$, we can conclude that $\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$. □

Since $\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$, we find that $P(2)$ is true.

Inductive Hypothesis: Assume $P(k)$ is true for $n = k$.

Inductive Step: We now show that $P(k+1)$ is true if $P(k)$ is true.

Under our inductive hypothesis, we assume that the following statement is true:

$$\overline{\bigcap_{j=1}^{k} A_j} = \bigcup_{j=1}^{k} \overline{A_j}$$

Using this, we want to show that $\overline{\bigcap_{j=1}^{k+1} A_j} = \bigcup_{j=1}^{k+1} \overline{A_j}$ is true. So, first, let us rewrite $\bigcup_{j=1}^{k+1} \overline{A_j}$ as $(\bigcup_{j=1}^{k} \overline{A_j}) \cup \overline{A_{k+1}}$. Next, let's use our inductive hypothesis to write $(\bigcup_{j=1}^{k} \overline{A_j}) \cup \overline{A_{k+1}} = (\overline{\bigcap_{j=1}^{k} A_j}) \cup \overline{A_{k+1}}$.

Now, we can use our base case, in which we proved that for sets S_1 and S_2, $\overline{S_1 \cap S_2} = \overline{S_1} \cup \overline{S_2}$. Setting $S_1 = \bigcap_{j=1}^{k} A_j$ and $S_2 = A_{k+1}$, this tells us that $(\overline{\bigcap_{j=1}^{k} A_j}) \cup \overline{A_{k+1}} = \overline{\bigcap_{j=1}^{k} A_j \cap A_{k+1}}$. Finally, we can rewrite this as $\overline{\bigcap_{j=1}^{k+1} A_j}$.

Therefore, we have shown that

$$\overline{\bigcap_{j=1}^{k+1} A_j} = \bigcup_{j=1}^{k+1} \overline{A_j}$$

Thus, if $P(k)$ is true, then $P(k+1)$ is true.

Conclusion:

We have shown that $P(2)$ is true, and if $P(k)$ is true, $P(k+1)$ is true for $k \geq 2$. Hence, we have demonstrated by induction that $P(n)$ is true for all $n \geq 2$.

Therefore, since our base case and inductive step are true, we can conclude that the generalization of DeMorgan's Law is true for all $n \geq 2$.

$\square$